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Context Columns

Enhance the bigram model with context columns that capture grammatical and semantic patterns.

You will need

- your completed bigram model from *Basic Training*
- pen, paper & dice as per *Basic Generation*

Your goal

To add new “context” columns to an existing bigram model and generate text from your newly context-aware model. **Stretch goal:** add and evaluate your own new context columns.

Key idea

The concept of attention — selectively focusing on relevant context — is a key innovation in Large Language Models. Adding context columns to your model gives it more information about which previous words matter most for prediction, leading to better generated text (with the trade-off being a slightly larger grid and more complex algorithm).



Algorithm (training)

1. **add context columns** to your existing bigram model:
after verb, *after pronoun* and *after preposition*
2. proceed as per *Basic Training*, but each time after updating the cell count for a word pair:
 - if the first word is a verb, increment the value in the second word's *after verb* column
 - if the first word is a pronoun (I/you/they etc.), increment the value in the second word's *after pronoun* column
 - if the first word is a preposition (in/on/at/with/to etc.), increment the value in the second word's *after preposition* column

This is a little tricky to get the hang of, but the key point is that you're updating two different rows each time — once for the “word follows word” cell, and once for the “context column” cell.

Example (training)

For text “I run fast. You run to me.” the model with context columns is:

	i	you	run	fast	to	me	.	after verb	after pronoun	after preposition
i										
you										
run										
fast										
to										
me										
.										

Algorithm (generation)

1. **choose a starting word**
2. check its row to identify the “normal” transition counts, but *also* check if the starting word is a verb/pronoun/preposition and if so add the values from the relevant “context” column before using a d10 to choose the next word
3. **repeat** from step 2 until you reach the desired length or a natural stopping point (e.g. a full stop .)

If you like, you can add your own context columns (based on patterns which *you* think are important).

Example (generation)

Starting word: **run** (a verb):

1. check **run** row: potential next words are **fast** (1) or **to** (1)
2. check all context columns: for **to** the **after verb** column has a count of 1 (appears after verbs)
3. combine both counts: roll a dice to choose either **fast** (1) or **to** (1 + 1 = 2)
4. repeat from step 1 until you reach the desired length or a natural stopping point (e.g. a full stop .)